



Paper id: 250608

Printed Page: 1 of 2
Subject Code: RCAI405

Roll No:

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MCA-INT
(SEM IV) THEORY EXAMINATION 2024-25
OPERATING SYSTEMS

TIME: 3 HRS**M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 7 = 14

a.	Write down the purpose of an operating system?
b.	What is meant by Interrupt?
c.	What is meant by Priority Scheduling?
d.	What are Pages and Frames?
e.	What is meant by Page Fault?
f.	What do you mean by a file?
g.	What is the difference between kernel and shell?

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

a.	What is an Operating System? Write down its desirable characteristics?
b.	What is semaphore? Discuss product-consumer problem with semaphore.
c.	What do you mean by paging in operating system? Explain.
d.	Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?
e.	Explain the functions of following UNIX commands: cat, cp, pwd, sort, grep.

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	What is distributed operating system? What are the advantages of distributed operating system?
(b)	What are the System Calls? Describe any four system calls.

4. Attempt any one part of the following:

7 x 1 = 7

(a)

What do you mean by PCB? Where is it used? What are its contents? Explain.

(b)

Consider the set of 6 processes whose arrival time and burst time are given below

Process Id	Arrival time	Burst time
P1	0	7
P2	1	5
P3	2	3
P4	3	1
P5	4	2
P6	5	1

If the CPU scheduling policy is shortest remaining time first, calculate the average waiting time and average turnaround time.



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M.MARKS:

5. Attempt any *one* part of the following:

7 x 1 = 7

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|-----|---|
| (a) | Consider the following page reference string: 1,2,3,4,5,3,4,1,2,7,8,7,8,9,7,8,9,5,4,5. How many page faults would occur for the following replacement algorithm, assuming four frames: a) FIFO b) LRU |
| (b) | Define a deadlock? Write down the conditions responsible for deadlock? How can we recover from deadlock? |

6. Attempt any *one* part of the following:

7 x 1 = 7

- | | |
|-----|--|
| (a) | Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?
a. FCFS b. SSTF c. SCAN |
| (b) | Discuss how free space is managed by operating system? |

7. Attempt any *one* part of the following:

7 x 1 = 7

- | | |
|-----|---|
| (a) | Explain the function of "Shell" in Unix Operating System. |
| (b) | Explain the Process Management in Linux. |